

Where Should the Adapter Go?

Outlier Dimensions, Rank Allocation and Placement in Low-Rank Adaptation of Small Language Models

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Abstract—Activation-guided methods such as EVA allocate a low-rank adapter’s capacity by profiling the target task’s activations beforehand. We test whether this helps small language models on biomedical classification at matched budgets, with each configuration tuned separately. Across eleven methods on three benchmarks, none beats a tuned uniform LoRA by more than 0.34 F1 (1.0 under HoC’s micro-F1), and no difference survives a multiple-comparison correction. A budget sweep, a 4M–110M ladder, a decoder SLM and a clinical task agree once multiple comparisons are corrected (1,328 runs). The methods differ in stability instead, and for activation-PCA initialisation we trace the cause. On RoBERTa-base, its leading directions lie on the backbone’s outlier dimensions 77 and 588 at 23–24 of the 24 distinct hidden-state inputs and, over all modules, carry a median $59\text{--}73\times$ an average direction’s energy, and budget-matched EVA loses 2 of 5 HoC seeds in fp16 and all 3 in fp32. The decoder’s signature is even stronger. Whitening these directions, deflating a general-domain reference subspace (DRIFT, which nests EVA) or the exact generalised-eigenvector contrast largely restore stability (whitening most reliably) but add no accuracy we can detect at this power. A random-token reference serves as well as real text, so the reference supplies the backbone’s outliers, not domain knowledge. Neither rank allocation nor placement helps at a matched budget, and halving the budget costs nothing, so capacity rarely binds; where it does, no allocation gain survives correction, but our power is lowest there. We close with a protocol for evaluating allocation.

Index Terms—parameter-efficient fine-tuning, low-rank adaptation, small language models, outlier dimensions, domain adaptation, biomedical text classification

I. INTRODUCTION

Text classification in specialised domains, such as clinical triage, pharmacovigilance and literature curation, is increasingly deployed under constraints that rule out large models: patient data cannot leave the institution, inference must run on commodity hardware, and dozens of task-specific classifiers must be served concurrently. Small language models (SLMs), from a few million to a few hundred million parameters, suit this setting, and parameter-efficient fine-tuning (PEFT) is the natural way to specialise them. A frozen backbone is shared across tasks, while a small adapter, typically around 1% of the parameters, carries the task-specific update [1], [2].

Given a fixed adapter budget, where in the network should the capacity go? Uniform allocation, as in standard LoRA [1], is budget-oblivious. One family of methods reallocates rank during training by importance scores [3]–[5]. A second, faster family does it *before* training, by profiling activations on a small calibration set. EVA initialises adapters from the principal directions of the target activations and redistributes rank by explained variance [6]. CorDA [7] and AIRA [8] weight the weight-matrix SVD by an activation covariance, and TLoRA [9] and Liu et al. [10] score layers by task sensitivity or effective rank.

Whether such allocation helps depends on how it is compared. Methods favour different learning rates, so a baseline that borrows the new method’s rate is not a neutral reference, and moving rank between modules of different width changes the number of parameters spent, so a matched rank is not a matched budget. Lee et al. [11] showed that LoRA and nine of its variants reach similar peak performance on decoder LLMs once every method’s learning rate is tuned. Training-free allocation has not been evaluated under the same protocol, nor in the regime where it should matter most, namely small backbones adapted to a specialised domain.

In domain adaptation the question takes a sharper form. Biomedical text is still English, so its dominant activation directions are largely those the pretrained model already handles, and profiling a single distribution may spend capacity where none is needed. Only a contrast between two distributions can reveal directions that carry target-domain energy *and* are poorly represented in pretraining. To give activation-guided allocation its best chance, we build that contrast into an instrument rather than a competing method. DRIFT (Domain-Residual Informed Fine-Tuning) deflates the target activation covariance by the principal subspace of a general-domain reference corpus, allocates a global parameter budget over the resulting *drift* spectrum, and initialises each adapter inside the residual subspace, with two forward passes and no labels or gradients. Its deflation level τ contains EVA as the special case $\tau = 0$, and its exact counterpart, the generalised eigenvectors of the target and reference covariances, tests the contrast in its

principled form.

Allocation should also matter most for small models: the intrinsic dimension of fine-tuning grows as models shrink [12], so a smaller backbone has less slack to absorb a misallocated budget, and we test this prediction on a controlled parameter ladder. The study asks four questions. **RQ1:** Under matched parameter budgets, with every configuration tuned on its own, does any allocation or initialisation scheme beat uniform LoRA? **RQ2:** Why is activation-PCA initialisation fragile, and does removing the cause help? **RQ3:** Which design choice matters (rank allocation, initialisation or placement), and does placement still matter when the budgets are matched? **RQ4:** Does the answer change with the backbone’s size or architecture?

Findings. This is an evaluation study, and its contributions lie in its findings. Under matched budgets and per-configuration tuning, no method beats LoRA by more than 0.34 F1, no difference with LoRA survives a Holm correction, and the methods differ in stability rather than accuracy. On RoBERTa-base, the leading directions of activation PCA lie on the backbone’s known outlier dimensions. Whitening, reference deflation and the generalised eigenvectors restore most of the stability, none turns it into accuracy, and random tokens serve as well as WikiText as the reference. Rank allocation alone slightly hurts, adapter placement changes nothing reliable at a matched budget, and half the budget matches the full one. On the BERT ladder no gain over LoRA survives correction, and EVA’s large deficit on small backbones at a rate tuned for RoBERTa-base shrinks to at most 2.5 points at its own rate. A decoder SLM and clinical notes repeat the pattern. Along the way, we provide DRIFT, a training-free two-distribution profiler that nests EVA, and a protocol for evaluating adapter allocation (Fig. 3).

II. RELATED WORK

Parameter-efficient fine-tuning and its evaluation. Adapter bottlenecks [13], prompt and prefix tuning [14], [15], bias tuning [16] and rescaling vectors [17] all restrict the update to a small subspace. LoRA [1] dominates because its update merges into the base weights; DoRA [18], VeRA [19] and QLoRA [20] refine it. Lee et al. [11] re-evaluate nine LoRA variants with a separate learning-rate search for each, find similar peak performance on decoder LLMs and trace the differences in their optimal rates to the largest eigenvalue of the loss Hessian, while Biderman et al. [21] document the gap that remains to full fine-tuning in domains far from pretraining. We extend the first kind of study to training-free rank allocation, matched parameter budgets and small encoders, and add a mechanistic account of the failure we observe.

Rank allocation and spectral initialisation. AdaLoRA [3] prunes singular triplets by a sensitivity score during training, SoRA [4] gates ranks with a sparsifying penalty and DyLoRA [5] trains nested ranks; all need a training run. Training-free allocation is more recent. EVA [6] redistributes rank by explained variance ratio, AIRA [8] by layer-wise

outlier ratios, TLoRA [9] by a sensitivity metric over $W\Sigma_x$, and Liu et al. [10] by effective rank plus a Fréchet distance under virtual perturbation, all from statistics of the target task alone. PiSSA [22] initialises from the leading singular directions of W , MiLoRA [23] from the trailing ones and LoRA-GA [24] from the first task gradient, and CorDA [7] orients the decomposition of W by an input covariance and, in its knowledge-preserving mode, uses a second distribution to decide what to freeze, but to our knowledge DRIFT is the first to *allocate rank* by a contrast between two distributions.

Where and at what scale to adapt. Which matrices to adapt goes back to LoRA itself [1, Sec. 7.1]. He et al. [25] find that modifying feed-forward representations uses added parameters more effectively than modifying attention. Our placement results test that finding under matched budgets and separately tuned learning rates. rsLoRA argues that $s = \alpha/r$ shrinks updates too fast as the rank grows and proposes $\alpha/\sqrt{r}$ [26], and LoRA+ gives A and B different learning rates [27]. We keep α/r , as EVA does, and tune the learning rate at every rank instead.

Domain adaptation and outlier dimensions. Continued pretraining [28] and domain pretraining [29]–[31] improve biomedical classification but update all parameters; we instead spend a small adapter budget on a frozen general-domain backbone. Contrasting the second-order statistics of two domains is classical: CORAL aligns covariances [32], subspace alignment aligns principal subspaces [33], and common spatial patterns take the generalised eigenvectors of two covariances [34], [35]. A handful of hidden dimensions in pretrained transformers carry very large activations, and disabling them badly damages BERT and RoBERTa [36]. For RoBERTa-base they are dimensions 77 and 588, whose magnitude tracks token frequency [37] and position [38]. Such rogue dimensions dominate similarity measures unless the representation is standardised [39], complicate quantisation [40], [41] and appear as massive activations in LLMs [42]. We show that they capture the leading directions of activation-PCA initialisation; the whitening we test is the adapter-space analogue of the standardisation remedy.

III. ACTIVATION-GUIDED ALLOCATION

Let the backbone contain adaptable linear modules $m = 1, \dots, M$ with weights $W_m \in \mathbb{R}^{d_m^{\text{out}} \times d_m^{\text{in}}}$. LoRA replaces W_m by $W_m + s B_m A_m$ with $A_m \in \mathbb{R}^{r_m \times d_m^{\text{in}}}$ and $B_m \in \mathbb{R}^{d_m^{\text{out}} \times r_m}$, so module m consumes $r_m c_m$ parameters with $c_m = d_m^{\text{in}} + d_m^{\text{out}}$. Given a global budget B we choose the ranks subject to $\sum_m r_m c_m \leq B$ and the initialisation of each A_m . When ranks are redistributed, we keep $s = \alpha/r$ for every module, as EVA’s reference implementation does.

A. Activation PCA and a reference contrast

For a linear module $y = Wx$, every gradient-descent update that starts from $\Delta W = 0$ has its row space in the span of the inputs x_i , so the input-side factor A should lie there, and a covariance argument tells us which r directions to pick.

Proposition 1 (Optimal rank- r input subspace): Let $\Delta W^* = \sum_{i=1}^N \delta_i x_i^\top$ with δ_i zero-mean, isotropic ($\mathbb{E}[\delta\delta^\top] = \sigma^2 I$) and independent of x_i , and let $P = A^\top A$ be the orthogonal projector onto the row space of a rank- r factor A . Then

$$\mathbb{E} \|\Delta W^*(I - P)\|_F^2 = \sigma^2 d^{\text{out}} N \text{tr}[(I - P) \Sigma_x], \quad (1)$$

with $\Sigma_x = \frac{1}{N} \sum_i x_i x_i^\top$, and the minimiser over rank- r projectors is the projector onto the top- r eigenvectors of Σ_x [43].

Proposition 1 justifies activation-PCA initialisation [6]. We apply it to the *centred* inputs, as EVA’s reference implementation does, since the mean contributes only a constant offset and in RoBERTa is dominated by a few outlier features [36], [37]. The proposition optimises the *total* gradient energy, however, which is dominated by directions where $\|x\|$ is simply large, and these need not be domain-specific.

Adaptation is driven by the *systematic* part of the gradient, $M = \mathbb{E}_D[\delta x^\top]$. For module m , let P_m^G be the orthogonal projector onto the top- k_m eigenspace of the covariance Σ_m^G of a general-domain corpus G (below we drop the index m where a single module is meant). We assume

$$(A) \quad \mathbb{E}_D[\delta (P^G x)^\top] \approx 0, \quad (2)$$

that is, the pretrained module is near-stationary on inputs confined to the reference subspace. (A) is a modelling assumption, not a derived result. It gives $\text{row}(M) \subseteq \text{range}(I - P^G)$, and applying Proposition 1 within that range makes the optimal input basis the leading eigenbasis of the *drift covariance*

$$\tilde{\Sigma} = (I - P^G) \Sigma^D (I - P^G). \quad (3)$$

No public corpus is the pretraining distribution, so (A) holds only approximately; we treat it as motivation and test its consequence (Section V-B).

B. One contrast, three approximations

Three ways of taming the target’s dominant directions appear in this paper, and all three approximate one operation: whitening the target covariance by a reference covariance R and taking its leading directions, i.e. solving $\Sigma^D v = \mu R v$. With $R = \Sigma^G$ the leading v maximise the ratio of target to reference energy. This is the criterion of common spatial patterns [34], [35] and of Fisher’s discriminant, and it suppresses directions that are energetic in both distributions, such as a backbone’s outlier dimensions. DRIFT’s deflation replaces $(\Sigma^G)^{-1}$ by a hard step, with zero weight on the reference’s top- k eigenspace and unit weight elsewhere. Whitened EVA keeps the target’s own principal directions and rescales each to the energy of an average direction, i.e. it whitens by the target covariance itself. We also run the exact criterion, **GEV**, in which each A_m holds the leading generalised eigenvectors of (Σ_m^D, Σ_m^G) , with Σ_m^G shrunk towards a scaled identity [44] and uniform rank, so that only the subspace differs from LoRA.

C. Budgeted allocation and initialisation

Let $\tilde{\lambda}_{m,1} \geq \tilde{\lambda}_{m,2} \geq \dots$ be the eigenvalues of $\tilde{\Sigma}_m$, and let $\hat{\lambda}_{m,i} = \tilde{\lambda}_{m,i} / \text{tr}(\Sigma_m^D)$ be the scale-free fraction of module m ’s target-domain energy that direction i captures outside the

reference subspace. We allocate by maximising total captured drift,

$$\max_{\{r_m\}} \sum_{m=1}^M \sum_{i=1}^{r_m} \hat{\lambda}_{m,i} \quad \text{s.t.} \quad \sum_{m=1}^M r_m c_m \leq B, \quad 0 \leq r_m \leq r_{\max}, \quad (4)$$

a separable concave resource-allocation problem that greedy marginal analysis solves to within the value of one rank unit (Proposition S1). Whether that objective tracks accuracy is the empirical question of Section V-C. P_m^G spans the fewest leading eigenvectors of Σ_m^G that capture a fraction τ of its trace; each module with $r_m > 0$ receives the leading eigenvectors of $\tilde{\Sigma}_m$ as A_m and $B_m = 0$, so the base function is preserved exactly. At $\tau = 0$ nothing is deflated and DRIFT is EVA, allocation by explained variance with initialisation from the principal directions [6]. This nesting attributes any difference from EVA to the deflation alone, and it is what makes DRIFT useful as an instrument even where it is not useful as a method (Section S-I).

IV. EXPERIMENTAL SETUP

Tasks. We use three biomedical benchmarks that differ in granularity, label structure and size. **ChemProt** is 13-way chemical–protein relation classification of PubMed sentences [45] in the split of Gururangan et al. [28], lexically the furthest from general English. **RCT-20k** is 5-way rhetorical-role classification of sentences in trial abstracts [46], whose signal lies in discourse structure rather than terminology. **HoC** is 10-label classification of abstracts by hallmark of cancer [47], rebuilt at document level so that it reproduces the BLURB split [29] (1,295 / 186 / 371 documents). HoC is scored with example-based F1 (micro-F1 where it allows a comparison with published numbers), the other two with micro-F1. A clinical task is introduced with its results (Section V-F), and Section S-II gives every split and setting.

Backbones. RoBERTa-base (125M) [48] is our primary backbone and lets us compare our numbers with published full fine-tuning results on the same splits [28]. For the scale study we use the BERT miniatures of Turc et al. [49], from BERT-Tiny (4.4M) to BERT-Base (110M), which share one vocabulary, pretraining corpus and recipe, so only depth and width vary. For the architecture check we use the decoder-only SmoLLM2-360M [50], classifying from its last token with a linear head and adapting all 224 linear modules.

Adapter placement and budget. All low-rank methods adapt every linear map in every transformer block, giving $M = 72$ modules for RoBERTa-base. The budget is that of uniform rank 8, $B = 1,327,104$ adapter parameters (1.07% of the backbone’s 124.06M). DRIFT and EVA redistribute rank but never exceed B . DoRA (1.41M) and AdaLoRA (up to 1.43M, after holding 1.5 $\times$ that during training) spend slightly more, which, if anything, favours them, and we report the parameters actually spent. Every method, including linear probing and BitFit, also trains the backbone’s classification head (0.60M parameters), which is identical across methods and excluded from the budget and from every parameter count we report.

DRIFT uses $\tau = 0.95$, a per-module cap $r_{\max} = 2r$ (as EVA) and relative scoring.

Methods. We compare full fine-tuning, linear probing, BitFit (all biases) [16], LoRA at uniform rank [1], DoRA [18], PiSSA [22] and AdaLoRA [3]. EVA [6] follows its reference code with one change: its greedy step spends our parameter budget B rather than a count of rank units, so that it competes under exactly the same budget. We call this variant *EVA (budget-matched)*, or EVA for short, and also run EVA’s own rank-unit rule. Because EVA proved unstable, we add its documented whitening option in a scale-normalised form, which rescales each principal row u_i by $\sqrt{\bar{\lambda}/\lambda_i}$ with $\bar{\lambda} = \text{tr}(\Sigma^D)/d^{\text{in}}$, so that only its *direction* differs from a random initialisation. Finally, we run DRIFT and GEV. Every low-rank method uses $\alpha = 16$, no adapter dropout, and, unless it prescribes its own initialisation, A drawn Kaiming-uniform and $B = 0$. All methods share one codebase (PyTorch 2.10, Transformers 5.0 [51]), data pipeline, optimiser and evaluation.

Training and tuning. We use AdamW [52] with linear warmup and decay, fp16 autocast with dynamic loss scaling (the T4 has no bf16), and select the epoch by dev-set score. Every configuration from which we draw a conclusion has its own learning rate, selected on the dev set with seed 1 from a grid that is extended by one half-decade step past whichever edge holds the best dev score until the selection is interior (Section S-XI lists all 119). This covers every method on every task, every LoRA placement and budget, every deflation level, every backbone of the ladder, the decoder, GEV and the clinical task, and variants that serve only as controls inherit their parent configuration’s rate. $\tau = 0.95$ was fixed before any experiment and never tuned. The test set is never used for any choice.

Reporting. We use three seeds, and five for LoRA, EVA, whitened EVA and DRIFT, the four methods on which the analysis turns, and for the reference controls. We report the mean and the sample standard deviation and compare methods with a paired t -test over shared seeds, correcting with Holm’s method [53] within each family of comparisons. A run is counted as *failed* when its best dev score is more than 10 points below the median of LoRA’s runs on the same task. The study comprises 1,328 fine-tuning runs, 443 of them for learning-rate selection, each on one NVIDIA T4.

Availability. All datasets are public. The code, the experiment plans, every per-run result file (with its per-epoch logs, where recorded) and the profile summaries are available at https://github.com/GowharYousuf07/drift_paper, together with the supplementary material, whose sections, tables and figures are numbered with an S. The clinical notes come from MTSamples, whose terms allow its sample reports to be shared for educational use with credit to the site. The Hugging Face release we use states no licence, so we redistribute no note text, and our code downloads it from that release.

V. RESULTS

A. RQ1: tuned LoRA is not beaten

Table I summarises the main results. Our training harness reproduces both published reference points available for these splits. Full fine-tuning of RoBERTa-base reaches 81.7 ± 0.4 micro-F1 on ChemProt, against 81.9 ± 1.0 reported by Gururangan et al. [28], and 80.0 ± 2.4 micro-F1 on HoC, against 79.7 reported for RoBERTa-base on the BLURB split [29].

No method beats uniform LoRA. At a matched budget of 1.07% of the backbone, no method improves on LoRA by more than 0.34 F1 on any task (BitFit on HoC, $p = 0.52$). After Holm correction over the 24 comparisons with LoRA, no difference in either direction is significant, and every adjusted p is 1.00. Only one difference reaches $p < 0.05$ uncorrected, and it is a loss: BitFit on ChemProt (-1.1 , $p = 0.04$). We read it, like every nominal result in this paper, as suggestive only. With three to five seeds, the seed-level tests rule out large gains but do not establish equivalence. Instance-level intervals from a replication of every parameter-efficient run (Section S-XIII) go further. On ChemProt and RCT-20k every interval ends below +1.2 points, so larger gains over LoRA are excluded, and 12 of the 16 lie within ± 2 points, which establishes equivalence at that margin (one we did not fix in advance); the other four extend past it only on the side of a loss. On HoC none ends above +3.4. GEV, the exact contrast, is within 0.3 points of LoRA on every task and fails no run.

Stability, not accuracy, separates the methods. HoC, with 1,295 training documents, 512-token inputs and ten labels, is where the methods differ, and the difference lies in failed runs. At its tuned learning rate, PiSSA fails in two of three seeds, DoRA in one, EVA in two of five and DRIFT in one of five, while LoRA, whitened EVA, AdaLoRA, BitFit and full fine-tuning never fail. Rerunning the column in fp32 at the same rates (Table II) separates two explanations. The four fp16 failures of PiSSA, DoRA and DRIFT disappear: none of their nine fp32 runs fails, including every seed that failed in fp16. EVA’s failures do not. All three of its fp32 seeds, including the two that trained in fp16, collapse early to predicting no label at all, through its initialisation or that initialisation’s interaction with the randomly initialised head (Section S-III). Across both precisions, the four initialisations line up with the median energy of their initial directions over all modules (Section V-B): whitened EVA (1 $\times$ by construction) fails nothing, DRIFT (1.5–2.7 $\times$) one fp16 run and no fp32 run, GEV (8–15 $\times$) no fp16 run and one fp32 run, and EVA (59–73 $\times$) two of five in fp16 and all three in fp32.

Selection at the edge of stability. On a task this small, single-seed learning-rate selection lands near the edge of stability, most often for methods that start from a data-derived subspace. Wherever we checked the alternative (EVA on HoC in fp32, AdaLoRA on RCT-20k and feed-forward-only rank 14 on HoC), selection by the mean over seeds picks the more stable rate. For EVA it removes its failures but leaves it 2.7 points below LoRA (Section S-III). We keep single-seed selection for every table entry, as Section IV specifies, and report the

TABLE I

TEST-SET RESULTS WITH A RoBERTa-BASE BACKBONE (MEAN $\pm$ S.D. OVER THREE SEEDS; $\dagger$ FIVE SEEDS). LOW-RANK METHODS ARE HELD TO THE ADAPTER-PARAMETER BUDGET OF UNIFORM RANK 8 (DoRA AND AdaLoRA EXCEED IT SLIGHTLY; THE COLUMN GIVES THE PARAMETERS ACTUALLY SPENT, EXCLUDING THE CLASSIFICATION HEAD THAT EVERY METHOD TRAINS). EVERY METHOD’S LEARNING RATE IS SELECTED ON THE DEV SET FROM A GRID EXTENDED UNTIL THE SELECTION IS INTERIOR (SECTION S-XI). HoC MED.: MEDIAN OVER SEEDS. FAILED: RUNS WHOSE BEST DEV SCORE IS MORE THAN 10 POINTS BELOW THE MEDIAN OF LoRA’S RUNS ON THE SAME TASK, OVER ALL THREE TASKS; NOT ASSESSED FOR THE LINEAR PROBE (–), WHICH CANNOT REACH LoRA’S LEVEL WITH A FROZEN BACKBONE. BEST PARAMETER-EFFICIENT RESULT PER COLUMN IN BOLD, TIES INCLUDED. BELOW THE RULE, THE TWO INSTRUMENTS OF SECTION III-B: DRIFT AND THE EXACT CONTRAST GEV, WHICH KEEPS UNIFORM RANK SO THAT ONLY ITS INITIALISATION DIFFERS FROM LoRA’S.

Method	Adapter params	ChemProt	RCT-20k	HoC	HoC med.	Failed
Full fine-tuning	124.06M	81.7 $\pm$ 0.4	83.7 $\pm$ 0.6	76.2 $\pm$ 1.9	75.5	0/9
Linear probe	0	49.3 $\pm$ 0.6	77.3 $\pm$ 0.6	48.8 $\pm$ 2.0	48.5	–
BitFit	0.10M	81.3 $\pm$ 0.1	83.9 $\pm$ 0.1	78.2$\pm$0.6	78.5	0/9
LoRA †	1.33M	82.5 $\pm$ 0.3	83.7 $\pm$ 0.3	78.0 $\pm$ 1.4	78.3	0/15
DoRA	1.41M	82.4 $\pm$ 0.6	83.6 $\pm$ 0.1	57.9 $\pm$ 37.8	79.1	1/9
PiSSA	1.33M	82.4 $\pm$ 0.6	83.8 $\pm$ 0.1	51.2 $\pm$ 33.0	61.5	2/9
AdaLoRA	1.42M	82.7$\pm$0.9	82.3 $\pm$ 1.8	78.1 $\pm$ 0.7	77.7	0/9
EVA (budget-matched) †	1.33M	81.5 $\pm$ 1.2	83.7 $\pm$ 0.2	59.4 $\pm$ 25.4	76.5	2/15
EVA (whitened) †	1.33M	82.0 $\pm$ 2.0	84.0$\pm$0.1	77.3 $\pm$ 1.3	77.8	0/15
DRIFT †	1.33M	82.4 $\pm$ 0.3	83.9 $\pm$ 0.1	75.7 $\pm$ 8.0	79.0	1/15
GEV	1.33M	82.4 $\pm$ 0.6	83.9 $\pm$ 0.1	77.6 $\pm$ 2.2	78.3	0/9

TABLE II

HoC IN fp16 (TABLE I) AND RERUN IN fp32 AT THE SAME LEARNING RATES. PyTorch’s DETERMINISTIC ALGORITHMS ON (EXAMPLE-BASED F1, THREE SEEDS; fp16 ROWS WITH † HAVE FIVE). FAILED: BEST DEV SCORE MORE THAN 10 POINTS BELOW LoRA’S MEDIAN; NOT ASSESSED FOR THE LINEAR PROBE (–).

Method	fp16			fp32		
	Mean	Med.	Failed	Mean	Med.	Failed
Full fine-tuning	76.2 $\pm$ 1.9	75.5	0/3	77.6 $\pm$ 1.2	77.3	0/3
Linear probe	48.8 $\pm$ 2.0	48.5	–	49.4 $\pm$ 2.0	49.1	–
BitFit	78.2 $\pm$ 0.6	78.5	0/3	78.9 $\pm$ 1.4	79.5	0/3
LoRA †	78.0 $\pm$ 1.4	78.3	0/5	78.9 $\pm$ 0.8	79.3	0/3
DoRA	57.9 $\pm$ 37.8	79.1	1/3	78.4 $\pm$ 0.2	78.4	0/3
PiSSA	51.2 $\pm$ 33.0	61.5	2/3	78.3 $\pm$ 0.8	78.2	0/3
AdaLoRA	78.1 $\pm$ 0.7	77.7	0/3	78.9 $\pm$ 0.9	78.4	0/3
EVA †	59.4 $\pm$ 25.4	76.5	2/5	14.3 $\pm$ 0.0	14.3	3/3
EVA (whitened) †	77.3 $\pm$ 1.3	77.8	0/5	76.7 $\pm$ 0.6	76.9	0/3
DRIFT †	75.7 $\pm$ 8.0	79.0	1/5	79.3 $\pm$ 0.7	79.2	0/3
GEV	77.6 $\pm$ 2.2	78.3	0/3	56.7 $\pm$ 36.7	76.6	1/3

resulting runs as they are instead of re-selecting after seeing test scores, but the protocol of Fig. 3 recommends the multi-seed rule.

EVA’s initialisation is the fragile part. Initialised along the target’s principal directions, EVA (budget-matched) trails LoRA on ChemProt with four times LoRA’s seed standard deviation (81.5 ± 1.2 vs. 82.5 ± 0.3 , five seeds) and falls to 59.4 ± 25.4 on HoC. Whitening its initial directions removes the HoC failures (77.3 ± 1.3 , none in five seeds). The problem thus lies in the energy of the chosen directions, not in the rank allocation, and Section V-B identifies what those directions are.

B. RQ2: why activation-PCA initialisation is fragile

For most RoBERTa-base modules, the leading principal direction of the input covariance is one of the backbone’s two known outlier dimensions rather than a direction of biomedical meaning (Fig. 1). In the modules that read the hidden state (W_Q , W_K , W_V and the feed-forward up-projection), the leading

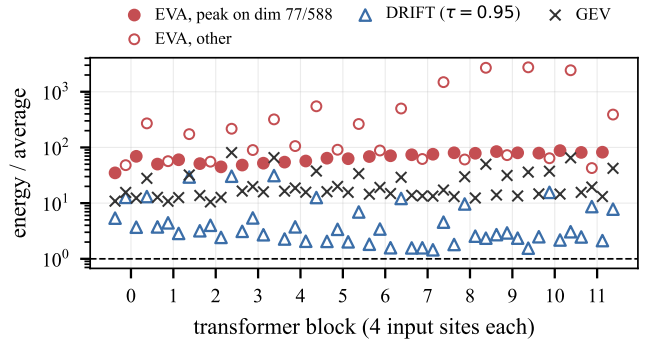


Fig. 1. Input energy along each module’s leading initial direction, relative to an average direction (log scale; RoBERTa-base, ChemProt). One point per input site: W_Q , W_K and W_V read the same block input. Filled: EVA’s leading direction peaks on outlier dimension 77 or 588; hollow circles: the attention-output and feed-forward hidden inputs, which are not in the hidden-state basis. Open triangles: DRIFT after deflation; crosses: GEV.

direction peaks on dimension 588 or 77. This happens at all 24 distinct hidden-state inputs on ChemProt and at 23 of 24 on RCT-20k and HoC. These directions carry a median 56–71 $\times$ the input energy of an average direction, and over all 72 modules the median is 59–73 $\times$, the aggregate we quote from here on unless we name a module group. An adapter whose input-side factor points along such a direction sees input energy one to three orders of magnitude larger than a randomly initialised one, so a learning rate that is comfortable for LoRA becomes a very large effective step: in the terms of [11], curvature rises and the usable learning rate falls. This matches three observations. EVA fails on HoC, where single-seed tuning selected LoRA’s rate for it; at ranks 1 and 2, where the fixed scale α/r is largest, it collapses already at 3×10^{-4} , whereas LoRA, whitened EVA and DRIFT still train at 10^{-3} ; and it recovers under whitening, which shrinks these rows. The selected learning rates are weaker evidence, since

they separate EVA from the low-energy initialisations only on ChemProt (Section S-IV).

The general-domain reference carries the same outlier dimensions, which are a property of the backbone and not of biomedical text, so the reference subspace absorbs them and deflation removes them: DRIFT’s leading drift directions carry a median of only 1.5–2.7 $\times$ average energy. What the reference removes first is therefore not “general English” in a linguistic sense, as our motivating argument assumed, but the backbone’s own idiosyncratic high-energy dimensions.

The reference corpus matters only through the backbone. With news articles, word-shuffled WikiText or uniformly random tokens as the reference, the reference subspace still absorbs the outlier dimensions, and the leading directions carry a median 3–4 $\times$ the energy of an average direction with news or shuffled text and 13–15 $\times$ with random tokens, against 1.5–2.7 $\times$ with WikiText. Accuracy follows none of these differences (Table III). No reference differs significantly from WikiText ($p \geq 0.07$), and the shuffled reference’s gain over LoRA on ChemProt (+0.6, $p = 0.0007$ uncorrected) rested on three nearly identical paired differences: over five seeds the gain shrinks to +0.4 ($p = 0.05$ uncorrected; $p = 0.24$ after Holm correction over the five reference-versus-LoRA comparisons). A reference made of random tokens is thus as good as real text: what the contrast contributes is the backbone’s outlier structure, which any input exposes, not knowledge of general English.

The exact contrast gives the same outcome. GEV’s leading directions avoid the outlier dimensions at all 24 hidden-state inputs and carry a median 8–15 $\times$ average energy (Fig. 1), and GEV is within 0.3 points of LoRA on every task ($p \geq 0.72$). Deflating even half of the reference energy removes what matters (Fig. S1): the robust effect of the whole family is the collapse of the seed scatter rather than a gain in the mean, and no member carries accuracy past LoRA.

C. RQ3: allocation or initialisation?

Table III factorises the design space. DRIFT changes two things at once, namely which modules get rank and which subspace each adapter starts in, so we vary the two independently on all three tasks. Combining DRIFT’s rank allocation with LoRA’s random initialisation scores lower than uniform LoRA in every seed on ChemProt (-0.3 , $p = 0.03$ uncorrected; like the other nominal results, suggestive only) and is destabilising on HoC, where one of its three seeds fails (65.0 ± 25.1). DRIFT’s initialisation with uniform ranks is the best low-rank configuration in the upper block on ChemProt (82.9 ± 0.9), though not significantly better than LoRA ($p = 0.46$), and it fails no run. Allocation is thus the half of the method that costs stability, while initialisation is the half that is free.

Allocation also has little to work with. If target and reference shared their geometry exactly, a fraction $1 - \tau$ of the target variance would fall outside the reference subspace by construction. Averaged over the 48 distinct input sites, the excess over $1 - \tau$ is only 2.5 (ChemProt), 2.2 (RCT-20k) and 1.4 (HoC) points of variance at $\tau = 0.95$, and

TABLE III

ABLATION (TEST F1, MEAN $\pm$ S.D.; SEEDS 1–3 THROUGHOUT, SO THE LoRA AND DRIFT ROWS DIFFER FROM THE FIVE-SEED VALUES OF TABLE I; 1.33M ADAPTER PARAMETERS EXCEPT EVA’S RANK-UNIT RULE, WHICH SPENDS 1.26–1.29M). OVER FIVE SEEDS THE REFERENCE ROWS GIVE CHEMPROT 82.3, 82.9, 82.6; HoC 78.0, 77.8 (SECTION V-B). UPPER BLOCK: ONLY THE ALLOCATION RULE AND THE INITIALISATION CHANGE, AT DRIFT’S LEARNING RATE. LOWER BLOCK: THE GENERALISED-EIGENVECTOR CONTRAST (OWN RATE), EVA WITH ITS OWN BUDGET RULE (EVA’S RATE), AND DRIFT WITH ITS WIKITEXT REFERENCE REPLACED. $\dashv$ NOT RUN.

Variant	ChemProt	RCT-20k	HoC
LoRA (uniform rank, random init)	82.5 $\pm$ 0.3	83.9 $\pm$ 0.2	77.9 $\pm$ 1.3
Rank allocation only (random init)	82.1 $\pm$ 0.3	83.9 $\pm$ 0.1	65.0 $\pm$ 25.1
Drift init only (uniform rank)	82.9 $\pm$ 0.9	83.9 $\pm$ 0.2	78.7 $\pm$ 1.1
Random orthonormal init, drift ranks	82.3 $\pm$ 1.0	–	–
No deflation ($\tau=0$), DRIFT’s rate	79.7 $\pm$ 2.3	–	–
Absolute (unnormalised) drift score	82.4 $\pm$ 0.6	–	–
DRIFT (full)	82.3 $\pm$ 0.4	84.0 $\pm$ 0.1	73.2 $\pm$ 10.2
GEV init (uniform rank)	82.4 $\pm$ 0.6	83.9 $\pm$ 0.1	77.6 $\pm$ 2.2
EVA, rank-unit budget (its own rule)	81.9 $\pm$ 0.5	–	78.4 $\pm$ 1.1
DRIFT, news reference	82.0 $\pm$ 0.2	–	77.8 $\pm$ 1.9
DRIFT, word-shuffled reference	83.1 $\pm$ 0.3	–	–
DRIFT, random-token reference	82.8 $\pm$ 0.4	–	78.4 $\pm$ 0.8

TABLE IV

ADAPTER PLACEMENT WITH LoRA. EACH CONFIGURATION AT ITS OWN TUNED LEARNING RATE (TEST F1, MEAN $\pm$ S.D., SEEDS 1–3; TABLE I REPORTS FIVE SEEDS FOR THE METHODS THAT HAVE THEM). RANK 14 ON THE FEED-FORWARD MATRICES SPENDS ABOUT THE ALL-MODULE RANK-8 BUDGET; RANK 4 ON ALL MODULES ABOUT THE FEED-FORWARD RANK-8 BUDGET. DRIFT ROWS ARE NOT TUNED: THEY INHERIT THE RATE OF ALL-MODULE DRIFT. TWO OF THREE SEEDS FAIL IN THE FEED-FORWARD RANK-14 HoC CELL AT ITS SELECTED RATE; AT THE RATE A THREE-SEED DEV MEAN SELECTS. ALL THREE TRAIN AND IT REACHES 77.0 $\pm$ 4.1 (SECTION V-D).

Placement	r	Params	ChemProt	RCT-20k	HoC
All modules	8	1.33M	82.5 $\pm$ 0.3	83.9 $\pm$ 0.2	77.9 $\pm$ 1.3
All modules	4	0.66M	82.6 $\pm$ 0.5	83.5 $\pm$ 0.5	78.7 $\pm$ 1.0
Feed-forward only	8	0.74M	83.4 $\pm$ 0.9	83.7 $\pm$ 0.2	78.0 $\pm$ 1.4
Feed-forward only	14	1.29M	82.8 $\pm$ 0.8	83.6 $\pm$ 0.4	35.0 $\pm$ 35.9
Attention only	8	0.59M	82.3 $\pm$ 0.5	83.9 $\pm$ 0.1	77.9 $\pm$ 3.6
DRIFT, feed-forward only	8	0.74M	82.8 $\pm$ 0.3	83.8 $\pm$ 0.3	75.6 $\pm$ 7.8
DRIFT, attention only	8	0.59M	81.2 $\pm$ 0.8	83.7 $\pm$ 0.1	78.2 $\pm$ 1.5

divergences that do not depend on τ agree (Section S-V): biomedical text largely reuses the geometry of general English. The visibly different allocation that DRIFT produces yields the same accuracy as uniform rank.

D. RQ3: where should the adapter go?

Allocation moves rank between modules that all carry an adapter, whereas placement decides which modules carry one at all. Adapting only the feed-forward matrices at rank 8 spends 0.74M parameters instead of 1.33M, so a comparison with all-module rank 8 confounds placement with budget. Table IV therefore compares each placement with another at the same budget, with every configuration tuned on its own: feed-forward-only rank 14 (1.29M) against all-module rank 8 (1.33M), and all-module rank 4 (0.66M) against feed-forward-only rank 8 (0.74M).

At matched budgets. Neither pair shows a reliable difference. Against all-module rank 8, feed-forward-only rank 14 is +0.3

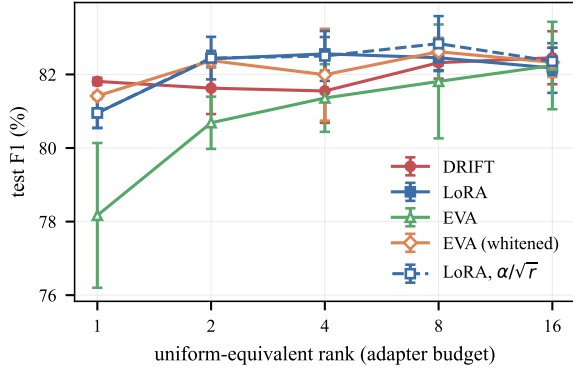


Fig. 2. Test micro-F1 against adapter budget (uniform-equivalent rank) on ChemProt, mean $\pm$ s.d. over three seeds, every (method, budget) pair at its own tuned learning rate (Section S-XI).

TABLE V

BACKBONE LADDER ON CHEMPROT (TEST MICRO-F1, MEAN $\pm$ S.D., THREE SEEDS): THE BERT MINIATURES SHARE ONE VOCABULARY AND PRETRAINING RECIPE. EACH METHOD RUNS AT THE LEARNING RATE TUNED ON THAT BACKBONE’S DEV SET; THE LAST COLUMN KEEPS EVA AT THE RATE TUNED FOR IT ON RoBERTa-BASE.

BERT	LoRA	EVA	EVA (whitened)	DRIFT	EVA (RoBERTa’s rate)
Tiny (4.4M)	60.9 $\pm$ 1.9	58.3 $\pm$ 1.1	59.3 $\pm$ 1.2	61.2$\pm$1.8	47.5 $\pm$ 3.4
Mini (11.2M)	70.8$\pm$0.9	68.4 $\pm$ 2.1	69.1 $\pm$ 0.8	70.1 $\pm$ 0.9	63.9 $\pm$ 0.8
Small (28.8M)	76.1$\pm$0.7	74.4 $\pm$ 0.4	74.6 $\pm$ 0.4	74.8 $\pm$ 0.7	71.1 $\pm$ 0.7
Medium (41.4M)	77.6$\pm$0.7	76.5 $\pm$ 1.3	76.7 $\pm$ 0.5	76.6 $\pm$ 1.1	74.9 $\pm$ 0.9
Base (110.1M)	79.2 $\pm$ 0.9	79.2 $\pm$ 1.3	79.8 $\pm$ 0.4	80.5$\pm$0.5	79.2 $\pm$ 1.3

on ChemProt ($p = 0.52$) and -0.2 on RCT-20k ($p = 0.53$). On HoC, two of its three seeds fail at the single-seed rate, and at the rate the three-seed dev mean selects it is 0.9 points below all-module rank 8 ($p = 0.78$) and 1.0 below feed-forward-only rank 8. Against feed-forward-only rank 8, all-module rank 4 is -0.8 on ChemProt ($p = 0.10$), -0.2 on RCT-20k ($p = 0.65$) and $+0.7$ on HoC ($p = 0.07$). The feed-forward advantage of He et al. [25] is therefore not a placement effect that survives budget matching here. Nor are fewer parameters an advantage: all-module rank 4 (0.66M) differs from all-module rank 8 by $+0.1$, -0.4 and $+0.8$ points ($p \geq 0.34$), and attention-only rank 8 (0.59M) by at most 0.2. At this budget and below it, capacity is not what limits accuracy. DRIFT placed on one module type, at the rate it inherits from all-module DRIFT, is never significantly better than LoRA placed there (Section S-V).

E. RQ4: budget, scale and architecture

Budget. Across budgets from rank 1 to 16 on ChemProt (Fig. 2), with the learning rate tuned at every budget, LoRA, DRIFT and whitened EVA stay within 1.0 points of each other at every budget, and LoRA is flat from rank 2 (82.4) to rank 16. Where the budget binds hardest, at rank 1, DRIFT has the best mean (81.8 ± 0.1 against LoRA’s 81.0 ± 0.4 ; $+0.9$, $p = 0.04$ uncorrected, not significant after Holm correction over the sweep’s twelve comparisons with LoRA). This is the

one place in our study where allocation might have room to help. On RCT-20k and HoC at ranks 4 and 16, the ranking matches Table I (Section S-XVI). HoC is the one task where the budget itself matters: LoRA at rank 16 beats LoRA at rank 8 by 2.4 points ($p = 0.05$ uncorrected), and no allocation method beats LoRA at that budget either.

Model scale. Table V tests the prediction that allocation should matter more as the backbone shrinks, with every method tuned on every backbone. The prediction is not borne out. LoRA has the best mean on BERT-Mini, -Small and -Medium, and DRIFT on BERT-Tiny ($+0.3$, $p = 0.89$) and BERT-Base ($+1.3$, $p = 0.03$ uncorrected), the largest gain over LoRA in the ladder but not significant after Holm correction over its twenty comparisons with LoRA. At their own tuned rates, EVA trails LoRA by 0.0 to 2.5 points and whitened EVA by at most 1.8, none significantly. The last column keeps EVA at the rate tuned for it on RoBERTa-base, as our first version of this study did. EVA then falls further behind as the backbone shrinks, to 2.7, 4.9, 6.9 and 13.4 points behind the tuned LoRA from BERT-Medium down to BERT-Tiny, and this column’s loss on BERT-Mini is the one difference in the ladder that survives correction. Most of that deficit comes from the inherited learning rate, the clearest illustration of why a rate tuned for one backbone is not a neutral choice for another. EVA’s remaining deficit grows as the backbone shrinks while its energy excess falls (Section S-VI), so it has another cause, and no allocation method’s difference from LoRA moves in its favour as the backbone shrinks, so the ladder gives the prediction no partial support either.

A decoder SLM. SmolLM2-360M has the strongest outlier signature we measured: one dimension hosts EVA’s leading direction at 41 of the 64 hidden-state inputs, consistent with the massive activations reported for LLMs [42], and EVA’s leading directions carry a median 125 $\times$ the energy of an average direction over all 224 modules. It reproduces the pattern seen on RoBERTa on ChemProt (Table S-I). No method beats LoRA (83.9 ± 0.5), and EVA is again the least stable (82.2 ± 3.6 , one seed at 78.1). On HoC the cause differs: in two of EVA’s three seeds the fp16 loss scale collapses. Rerun in fp32, EVA trains in every seed without a non-finite step. It scores 79.8 ± 1.7 , level with LoRA’s 80.7 ± 1.5 in fp32 (-0.9 , $p = 0.66$), so on the decoder EVA’s instability is fp16 overflow rather than an early collapse, and removing it again adds no accuracy.

F. Clinical notes

To test whether the findings survive the shift towards clinical style, we classify transcribed clinical notes from MT-Samples [54] into 12 specialties (Section S-VII). Both the mechanism and the conclusions carry over. EVA’s leading directions peak on dimension 77 or 588 at 21 of the 24 hidden-state inputs and carry a median 57 $\times$ the energy of an average direction over all 72 modules, compared with 24 of 24 and 73 $\times$ on ChemProt, and deflation leaves one such input and a median of 4.0 $\times$. LoRA is again unbeaten—84.6 micro-F1 (84.3 macro) against DRIFT’s 84.5 (-0.1 , $p = 0.86$),

A protocol for evaluating adapter allocation

- 1) *Budget in parameters.* Match adapter parameters, not total rank, and report the parameters actually spent (head excluded).
- 2) *Tune every configuration.* Select the learning rate separately for every method, backbone, budget and placement, by the mean dev score over at least three seeds (on small tasks one seed can land on the edge of stability, Section V-A), and extend each grid until the selection is interior.
- 3) *Seeds and failures.* Use at least five seeds on small tasks. Report medians and failed runs next to means, rerun failures in fp32 to separate numerics from method, and correct for multiple comparisons.
- 4) *Energy check.* Before training a data-derived initialisation, compute each initial direction’s energy relative to an average direction (target forward pass plus one leading eigenvector per module). On RoBERTa-base and SmolLM2-360M, medians of 59–125 $\times$ coincided with failed or unstable runs and 1–15 $\times$ with stable training; on the BERT ladder 20–59 $\times$ trained stably at tuned rates. Read a high median as a warning to lower the rate, whiten or deflate—not as a verdict.
- 5) *Check that the budget binds.* Before trying allocation, run LoRA at half the budget and at other placements of the same budget. If nothing changes, allocation has nothing to exploit. Neither check needs profiling.

Fig. 3. A protocol for evaluating adapter allocation, distilled from Section V.

EVA’s 83.5 (−1.1, $p = 0.20$) and whitened EVA’s 83.4 (−1.2, $p = 0.13$)—and the random-token reference serves as well as WikiText (Table S-II). Profiling itself costs less than half a LoRA training run, its reference pass can be reused across domains, and it adds no inference cost, since the adapter merges into the base weights (Section S-VIII).

VI. DISCUSSION: EVALUATING ADAPTER ALLOCATION

Inference cost does not separate the methods we compare. Every low-rank adapter merges into the backbone after training, so the costs that differ are training and profiling time, and the only question is whether profiling buys accuracy. Our results do not say that adapter capacity is irrelevant, since at rank 1 it does bind. What they say is that, for small encoders at a budget of about 1%, where the capacity goes (across modules or across module types) matters less than how each configuration is trained, that is, the learning rate selected for it and the energy of its initial directions. Fig. 3 condenses these lessons into a protocol, and Section S-IX gives the evidence behind each step. Our results support the defaults that PEFT libraries ship (adapters on all linear maps, $s = \alpha/r$, Kaiming-uniform A and $B = 0$) for small encoders, with two amendments: report the parameters an adapter actually spends, and make data-derived initialisations report the energy of their directions.

VII. LIMITATIONS

Three to five seeds detect only large differences: the instance-level intervals exclude gains above 1.2 points on ChemProt and RCT-20k, but on HoC only those above 3.4.

Learning rates are selected on one seed, which on HoC puts several methods near the edge of stability, and the controls of Table III inherit their parent configuration’s rate. The MTSamples notes are sample reports published for medical transcriptionists, not records from an electronic health record, so they test clinical style rather than clinical deployment, and because every profile-based method centres its inputs, a mean shift between domains is invisible to them by construction. Input-side profiling cannot separate W_Q , W_K and W_V , which share one block input; WikiText is only a proxy for the pre-training distribution; architectures without pronounced outlier dimensions may not show EVA’s instability; and larger decoders, generative formulations and other specialised domains remain untested (Section S-X).

VIII. CONCLUSION

For practitioners the advice is short. On small encoders at budgets like ours, use all-module LoRA at rank 4 to 8, select its learning rate by the mean dev score of at least three seeds, check whether a larger budget helps before trying to allocate one, and skip data-derived allocation. If you do use a data-derived initialisation, check the energy of its directions and whiten them when it is high.

Activation-guided allocation asks where the target distribution’s variance is largest. On RoBERTa-base, for most modules, the answer is one of two outlier dimensions of the backbone, and adapters initialised there are fragile. Whitening those directions, deflating them with a general-domain reference (DRIFT), or taking the generalised eigenvectors of the target and reference covariances restores most of the stability without labels, gradients or inference cost. But under a matched parameter budget, with every configuration tuned on its own, no such fix and no allocation rule we tested outperformed uniform LoRA on three biomedical benchmarks, a clinical task or a decoder SLM. Rank allocation alone slightly hurt, adapter placement changed nothing reliable at a matched budget, and half the budget was enough. This verdict holds where the budget does not bind, which at the budgets we tested is almost everywhere. Where it binds (rank 1 on ChemProt, and HoC, where rank 16 beats rank 8), no allocation rule gained after correction, but three seeds give us the least power there, and we tested only input-side, activation-derived signals. On smaller backbones the learning rate tuned for each backbone, not any allocation rule, decided how far EVA trailed LoRA.

We believe this is the right baseline for the field to report against. Improvements claimed for activation-aware allocation should be measured at matched parameters, against a separately tuned LoRA, at learning rates tuned for the backbone, budget and placement in question, over enough seeds to reveal their failures, and checked for the stability of their initialisation. Natural next steps are to test the outlier mechanism in larger decoders, generative tasks and real clinical records, and to ask whether a cheap output-side signal can make allocation matter where input geometry alone does not.

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